La vogue vista: a cross-platform AI-powered mobile platform for real-time virtual beauty try-on and personalized recommendation

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Abstract. La Vogue Vista is a cross-platform mobile application that improves digital beauty product evaluation through real-time augmented reality visualization and AI-assisted recommendation. The system combines Flutter for user workflows, Firebase for authentication and catalog persistence, and native iOS/Android rendering for low-latency camera processing. MediaPipe landmark tracking and TensorFlow Lite segmentation align lipstick and hair overlays while preserving texture continuity. A cloud-assisted module, using OpenRouter-based language reasoning over structured visual descriptors, provides explainable style and product suggestions. The architecture follows a hybrid strategy: latency-critical tracking and rendering run on-device, while higher-level semantic recommendation uses cloud inference. Device-level implementation tests show interactive operation for live lipstick and hair try-on, with practical recommendation support for product exploration.

Keywords: Artificial Intelligence, Augmented Reality, Computer Vision, Edge AI, Mobile Commerce

1 Introduction

E-commerce beauty selection is difficult because 2D product swatches do not reliably represent appearance across skin tones, facial geometry, and lighting conditions. This mismatch increases uncertainty and weakens purchase confidence.

Existing solutions are often fragmented, where one application focuses on makeup while another handles hair simulation or recommendation. This work proposes a unified mobile platform that integrates real-time AR try-on and AI-supported personalization in one workflow.

2 Literature review

Recent AR beauty systems rely on facial landmarks and semantic segmentation. Landmark methods provide efficient geometric anchoring, while segmentation supports pixel-level recoloring. MediaPipe and TensorFlow Lite are commonly used in mobile deployments because they balance speed and inference quality.

Recommendation systems increasingly use language-model reasoning for explainable guidance. However, many pipelines separate visual simulation from semantic recommendation. This project adopts a hybrid architecture that unifies both.

3 Methodology

3.1 System architecture

The system uses Flutter UI orchestration, Firebase Auth/Firestore/Storage, native Swift/Kotlin camera rendering layers, MediaPipe plus TFLite for on-device computer vision, and OpenRouter LLM inference for explainable recommendations.

3.2 Geometric AR formulation (makeup and hair)

Given normalized landmarks p_i=(x_i, y_i), region alignment is computed by center, orientation, and scale.

$$c = (1/N) * \mathrm{sum}(p_i),\ \theta = \mathrm{atan2}(y_b - y_a, x_b - x_a),\ s = \|p_b - p_a\| / d_0 \quad (1)$$

Overlay transform:

$$T = \mathrm{Translate}(c) * \mathrm{Rotate}(\theta) * \mathrm{Scale}(s) \quad (2)$$

Hair mask compositing:

$$I_{out} = (1 - \alpha m) * I_{in} + \alpha m * I_{shade} \quad (3)$$

Luminance control:

$$L_{out} = \max(L_{in}, \lambda * L_{target}) \quad (4)$$

3.3 Recommendation inference

Structured descriptors (face geometry, tone features, hair attributes) are transformed into prompt-safe metadata and submitted to cloud LLM inference for explainable style and product guidance.

4 Results and discussion

The implemented system supports integrated shopping and try-on workflows with interactive behavior on physical devices for lipstick and hair try-on. User interaction remains responsive under typical indoor lighting and moderate pose variation.

The hybrid architecture reduces visual latency by retaining geometric tracking and rendering on-device, while cloud reasoning improves recommendation richness. Observed trade-offs include sensitivity to illumination, camera quality, and segmentation confidence during rapid motion.

5 Conclusion and recommendation

La Vogue Vista demonstrates that a unified mobile platform combining AR visualization and AI recommendation is technically feasible and practically useful for beauty product evaluation. Future work includes broader device benchmarking, controlled user studies, and robustness improvements for challenging real-world conditions.

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